Ezan Khan

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EDUCATION

Boston University | Boston, MA — May 2026
Bachelor of Science in Electrical Engineering, Minor in Computer Science — GPA: 3.53
Coursework: Software Engineering, Web Development, Computer Networking, Analysis of Algorithms, Digital Logic Design, Signals and Systems, Analog Electronics, Power Electronics, Electric Energy Systems

PROJECTS

Treebot | *Python, Raspberry Pi, Flask, Motor Control, Power Electronics* — May 2026

- Designed a web-controlled robot that climbs 6–12 inch diameter tree trunks using dual linear actuators and servo-driven gripping arms, reaching heights up to 20 feet while supporting payloads up to 50 lbs.
- Developed a Wi-Fi-based control system with live video streaming and real-time sensor feedback, enabling remote climbing, arm actuation, chainsaw on/off control, and horizontal chainsaw positioning from up to 100 feet away.
- Designed a 120V AC-powered electrical system with voltage regulation, motor drivers, and fail-safe protection, validating subsystem performance using oscilloscopes and multimeters.

New England Power Grid Modeling & Optimization | *MATLAB, Power Systems, Convex Optimization* — Dec 2025

- Developed a data-driven load forecasting model using convex optimization to predict AI-driven electricity demand growth in ISO New England through 2035.
- Simulated grid behavior using DC Optimal Power Flow (DC OPF) on the IEEE-39 bus system, analyzing congestion, locational marginal prices (LMPs), and system cost under high-load scenarios.
- Designed and evaluated generation expansion strategies (repowering and new generation placement), restoring network feasibility while reducing congestion and quantifying tradeoffs in monetary cost and carbon emissions.

LEGO x Minecraft Modular Lamp System | *C++, Power System Design, Housing Design, Woodwork* — Feb 2025

- Built a physical recreation of Minecraft redstone logic where a removable torch triggers different outputs based on placement, such as actuating a servo-driven piston or activating a 1000-lumen lamp.
- Integrated mechanical, electrical, and LEGO-inspired modular housing components to ensure stable actuation and repeatable switching behavior.

EXPERIENCE

Office Assistant | Boston University Romance Studies Department — Boston, MA | Sep 2022 - May 2026

- Greeted visitors, answered phone calls, and managed department mail and parcels.
- Helped set up large, department-wide events, including book sales and holiday parties.

Calculus Grader | Boston University Math Department — Boston, MA | Sep 2023 - Dec 2025

- Graded weekly Calculus quizzes and problem sets for students using self-developed grading rubrics.

Embedded Systems Research Assistant | Forlano Lab — Brooklyn, NY | May 2024 - Aug 2024

- Built an underwater acoustic sound trap with speakers and microphones to record toadfish calls.
- Developed MATLAB scripts to process hours of acoustic recordings to classify toadfish call patterns.
- Performed digital traces on fish tissue, resulting in co-author credit on research presented at the 2025 Society for Integrative and Comparative Biology (SICB) Conference.

SKILLS & AWARDS

Software: React, Flask, REST APIs, JSON, JUnit, Android Studio
Languages: MATLAB, Python, C++, Java, JavaScript, HTML, CSS, R
Tools: Git, GitHub, VS Code, AutoCAD, LTspice, KiCad, Fusion 360, Onshape, Tinkercad
Hardware: Soldering, PCB Design, Motor Control, CNC Machining, 3D Printing, Waterjet Cutting
Awards: BU ECE Senior Design Best PPE Award (Dec 2025), Lutron Lighting Innovation Winner (Feb 2025), IAC Charitable Foundation Scholarship (Nov 2024), BU Student Employee of the Year Nominee (Mar 2024)